The documentation and process conversion measures necessary to comply with this revision shall be completed by 4 November 2006.

INCH-POUND

MIL-PRF-19500/647C
4 August 2006
SUPERSEDING
MIL-PRF-19500/647B
24 October 2005

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER,
DUAL, ULTRAFAST, TYPES 1N6778 AND 1N6779,
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 This specification covers the performance requirements for silicon, ultrafast, power rectifier diodes in a center-tap configuration. Four levels of product assurance are provided for oath device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (2 pin, isolated - TO-257).

1.3 Maximurn ratings.

| Types | $V_{RWM}$ (1) $I_D$ = 10 µA dc | $I_F$ (1) (2) $T_C$ =+100°C | $I_{FSM}$ (1) $t_p$= 8.3 ms | $R_{\theta JC}$ (1) | $R_{\theta JA}$ (1) | $T_{STG}$ and $T_J$ |
|---|---|---|---|---|---|---|
| | Vdc | A dc | A (pk) | °C/W | °C/W | °C |
| 1N6778<br>1N6779 | 400<br>600 | 15 | 140 | 1.8 | 40 | -65 to +50 |

(1) Each individual diode.
(2) Derate at 300 mA/°C above $T_C$ = +100°C.

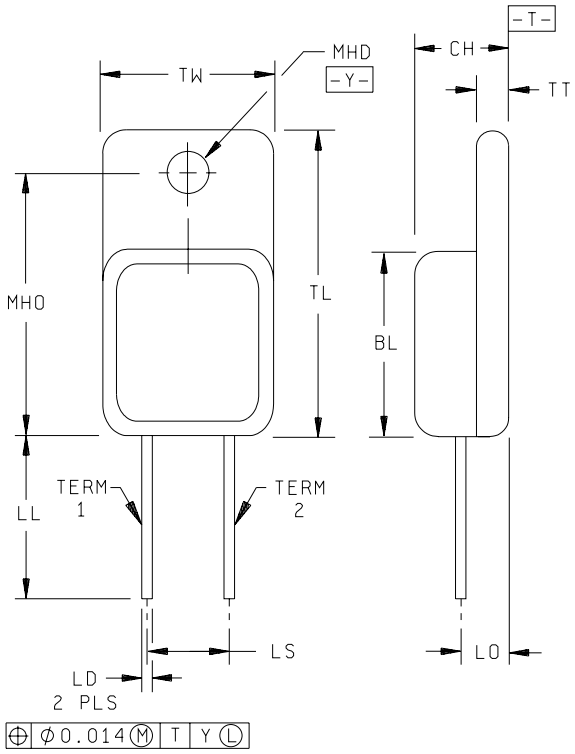
1.4 Primary electrical characteristics. Unless otherwise specified, primary electrical characteristics are at +25°C, and for each diode.

| Types | $V_{F1}$ $I_F$ = 8 A dc | $V_{F2}$ $I_F$ =15 A dc | $I_{R1}$ (see 1.3) $V_R$ = 0.8 $V_{RWM}$ | $I_{R2}$ $V_R$ = 0.8 $V_{RWM}$ (see 1.3) $T_C$ = +100°C | $t_{rr}$ | $C_J$ $V_R$ = 5 V f = 1 MHz |
|---|---|---|---|---|---|---|
| | V dc | V dc | µA dc | µA dc | ns | pF |
| 1N6778<br>1N6779 | 1.40 | 1.60 | 10 | 1,000 | 60 | 300 |

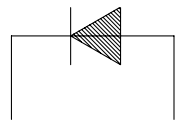
Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

AMSC N/A FSC 5961

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SCHEMATIC



1 2

Configuration

| Terminal | Description |
|---|---|
| 1 | Cathode |
| 2 | Anode |

* FIGURE 1. Physical dimensions and configuration (2 pin, isolated) (TO-257).



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| Dimensions | | | | |
|---|---|---|---|---|
| Symbol | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .410 | .430 | 10.4 | 10.9 |
| CH | .249 | .260 | 6.32 | 6.60 |
| LD | .035 | .045 | 0.89 | 1.14 |
| LL | .500 | .750 | 12.70 | 19.05 |
| LO | .150 typ | | 3.81 typ | |
| LS | .200 bsc | | 5.08 bsc | |
| MHD | .140 | .150 | 3.56 | 3.81 |
| MHO | .527 | .537 | 13.4 | 13.6 |
| TL | .645 | .665 | 16.4 | 16.9 |
| TT | .040 | .050 | 1.02 | 1.27 |
| TW | .410 | .420 | 10.4 | 10.7 |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. All terminals are isolated from case.
4 In accordance with ASME Y14.5M, diameters are equivalent to $\phi$x symbology.

* FIGURE 1. Physical dimensions and configuration (2 pin, isolated) (TO-257) - Continued.



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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil. or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Polarity. Polarity and terminal configuration shall be in accordance with figure 1 herein.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be group A as specified herein.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.



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3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.3 Screening (JANTX, JANTXV, and JANS levels). Screening shall be in accordance with appendix E, table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see appendix E, table E-IV of MIL-PRF-19500) | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| (1) 3c | Thermal impedance (see 4.3.2) | Thermal impedance (see 4.3.2) |
| 7a and 7b | Optional | Optional |
| 9 and 10 | Not applicable | Not applicable |
| 11 | $I_{R1}$ and $V_{F1}$ | $I_{R1}$ and $V_{F1}$ |
| 12 | See 4.3.1, t = 240 hours | See 4.3.1, t = 48 hours |
| 13 | Subgroups 2 and 3 of table I herein; $V_{F1}$ and $I_{R1}$; $\Delta I_{R1} \leq$ 100 percent of initial value or ±2.5 μA, whichever is greater; $\Delta V_{F1} \leq$ ±100 mV. | Subgroup 2 of table I herein; $V_{F1}$ and $I_{R1}$; $\Delta I_{R1} \leq$ 100 percent of initial value or ±2.5 μA whichever is greater; $\Delta V_{F1} \leq$ ±100 mV. |
| 14a and 14b | Required | Required |

(1) Thermal impedance shall be performed any time after temperature cycling, screen 3a, and does not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: Method 1038 of MIL-STD-750, test condition A. $T_C$ = +125°C; $V_R$ = 0.8 of rated $V_{RWM}$ (see 1.3).

* 4.3.2 Thermal impedance The thermal impedance measurements shall be performed in accordance with method 3101 or 4081 of MIL-STD-750, using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$, ($V_C$ and $V_H$ where appropriate). Measurement delay time ($t_{MD}$) = 35 μs max. See table II, subgroup 4 herein.



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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I, herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table III herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VIa (JANS) and table E-VIb (JANTX and JANTXV) of MIL-PRF-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.2.1 Group B inspection, appendix E, table E-VIa (JANS) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B4 | 1037 | $I_F$ or $I_O$ = 1.25 A to 10 A; $\Delta T_J$ = +85°C minimum, for 2,000 cycles minimum. |

4.4.2.2 Group B inspection, appendix E, table E-VIb (JANTX and JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1037 | $I_F$ or $I_O$ = 1.25 A to 10 A; $\Delta T_J$ = +85°C minimum, for 2,000 cycles minimum. |
| B5 | | Not applicable |

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table III herein.

* 4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition A, 5 pounds, t = 15 seconds ±3 seconds. |
| C5 | 4081 | See 4.3.2. |
| C6 | 1037 | $I_F$ or $I_O$ = 1.25 A to 10 A; $\Delta T_J$ = +85°C minimum, for 6,000 cycles minimum. |

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and table II herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps and footnotes of table III herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.



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* TABLE I. Group A inspection. 1/ 2/

| Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance | 3101 | See 4.3.2 | $Z_{\theta JX}$ | | | °C/W |
| Breakdown voltage<br>1N6778<br>1N6779 | 4022 | $I_R$ = 10 µA dc, pulsed 3/ | $V_{BR}$ | <br>400<br>600 | | V dc |
| Forward voltage | 4011 | $I_F$ = 8 A dc, pulsed 3/<br>$I_F$ = 15 A dc, pulsed 3/ | $V_{F1}$<br>$V_{F2}$ | | 1.40<br>1.60 | V dc<br>V dc |
| Reverse leakage current | 4016 | DC method; pulsed 3/<br>$V_R$ = 0.8 of $V_{RWM}$ (see 1.3) | $I_{R1}$ | | 10 | µA dc |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_C$ = +100°C | | | | |
| Reverse leakage Current | 4016 | DC method; pulsed 3/<br>$V_R$ = 0.8 of $V_{RWM}$ (see 1.3) | $I_{R2}$ | | 1.0 | µA dc |
| Low temperature operation: | | $T_A$ = -55°C | | | | |
| Forward voltage | 4011 | $I_F$ = 15 A dc, pulsed 3/ | $V_{F3}$ | | 1.80 | V dc |
| Subgroup 4 | | | | | | |
| Scope display evaluation 4/ | 4023 | | | | | |
| Reverse recovery time measurements | 4031 | Condition B1<br>$I_F$ = 1.0 A,<br>di/dt = 50 A/µs | $t_{rr}$ | | 60 | ns |

See footnotes at end of table.



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* TABLE I. Group A inspection – Continued. 1/ 2/

| Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroups 5 and 6 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 7 | | | | | | |
| Junction capacitance | 4001 | $V_R$ = 5 V dc; f = 1.0 MHz | $C_J$ | | 300 | pF |

1/ For sampling plan, see MIL-PRF-19500.
2/ Each individual diode.
3/ Pulse test: Pulse width = 300 μs, duty cycle ≤ 2 percent.
4/ The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 50 to 100 μA/division and 50 to 100 V/division. Reverse current over the knee shall be at least 500 μA. Each device may exhibit a slightly rounded characteristic and any discontinuity or dynamic instability of the trace shall be cause for rejection.



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* TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

| Inspection | MIL-STD-750 | | Sampling plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 22 devices<br>c = 0 |
| Temperature cycling | 1051 | 500 cycles | |
| Hermetic seal | 1071 | | |
| Fine leak | | Test condition H<br>5 x $10^{-7}$ atm cc/s | |
| Gross leak | | Test condition C or K | |
| Electrical measurements | | See table III herein, steps 1, 2, 3, and 5 | |
| Subgroup 2 | | | 22 devices<br>c = 0 |
| Steady-state reverse bias | 1038 | Test condition A<br>t = 1,000 hours, $T_C$ = +125°C<br>$V_R$ = 0.8 of rated $V_{RWM}$ (see 1.3) | |
| Electrical measurements | | See table III herein, steps 1 and 2 | |
| Subgroup 4 | | | |
| Thermal impedance curves 1/ 2/ 3/ | | See MIL-PRF-19500. | |
| Subgroup 5 | | | |
| Barometric pressure, reduced (altitude operation) | 1001 | Pressure 8.0 mm | 15 devices<br>c = 0 |

1/ The approved thermal impedance conditions and limit for $Z_{\theta JX}$ shall be used by the supplier in screening and table I, subgroup 2.

2/ The approved thermal resistance conditions for $R_{\theta JX}$ shall be used by the supplier for conformance inspection.

3/ For product families with similar thermal characteristics based on the same physical and thermal die, package, and construction combination (thermal grouping), the supplier may use the same thermal impedance curves.



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* TABLE III. Groups A, B, C, and E electrical and delta measurements. 1/ 2/ 3/ 4/

| Step | Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1 | Forward voltage | 4011 | $I_F$ = 15 A dc pulsed | $V_{F2}$ | | 1.60 | V dc |
| 2 | Reverse leakage current | 4016 | $V_R$ = 0.8 of percent $V_{RWM}$ (see 1.3) DC method, pulsed | $I_{R1}$ | | 10 | μA dc |
| 3 | Forward voltage | 4011 | $I_F$ = 15 A dc pulsed | $\Delta V_{F2}$ | | ±100 mV dc from initial reading | |
| 4 | Reverse leakage current | 4016 | $V_R$ = 0.8 of percent $V_{RWM}$ (see 1.3) DC method, pulsed | $\Delta I_{R1}$ | | 100 percent of initial value or ±2.5 μA dc whichever is greater. | |
| 5 | Thermal impedance | 3101 | See 4.3.2 | $Z_{\theta JX}$ | | 1.6 | °C/W |

1/ The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:
- a. Subgroup 3, see table III herein, steps 1 and 2.
- b. Subgroup 4, see table III herein, steps 1, 2, 3, 4, and 5.
- c. Subgroup 5, see table III herein, steps 1, 2, 3, and 4.

2/ The electrical measurements for table VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows:
- a. Subgroup 2, see table III herein, steps 1, 2, and 5.
- b. Subgroup 3, see table III herein, steps 1, 2, 3, 4, and 5.
- c. Subgroup 6, see table III herein, steps 1 and 2.

3/ The electrical measurements for table VII of MIL-PRF-19500 are as follows:
- a. Subgroup 2, see table III herein, steps 1, 2, and 5 for all levels.
- b. Subgroup 3, see table III herein, steps 1 and 2.
- c. Subgroup 6, see table III herein, steps 1, 2, 3, 4, and 5 for all levels.

4/ The electrical measurements for table IX of MIL-PRF-19500 are as follows:
- a. Subgroup 1, see table III herein, steps 1, 2, 3, and 5.
- b. Subgroup 2, see table III herein, steps 1 and 2.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Cross reference list. Parts covered by this specification may be used to replace the following commercial Part or Identifying Numbers (PIN):

| Preferred types | Commercial types |
|---|---|
| JAN1N6778, JANTX1N6778, JANTXV1N6778,JANS1N6778<br>JAN1N6779, JANTX1N6779, JANTXV1N6779,JANS1N6779 | 1N6778<br>1N6779 |



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6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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